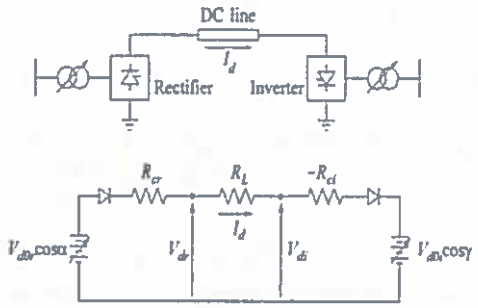
EE4-53

Part A: Answer any 2 questions out of 3 from this part

1. a) Sketch the equivalent circuit of a point-to-point LCC HVDC link and explain what each component on either side represents.

[5]



The equivalent circuit comprises of two controllable voltage sources, the commutation resistances and the line resistance. The diodes signify that the current can only flow in one direction. Controllable voltage sources at either end depend on the AC system voltage and the firing angles. The commutation resistances represent the drop (reduction) in terminal voltage due to commutation overlap.

V_{dor}/V_{doi} – no-load ideal direct voltage at rectifier/inverter end

V_{dr}/V_{di} – voltage at rectifier/inverter end

R_{cr}/R_{ci} – commutation resistance at rectifier/inverter end

R_L – line resistance

α/γ – firing/firing advance angle at rectifier/inverter end

I_d – line current

[2 marks for proper sketch and labelling; 3 marks for explaining the components]

b) Explain the problem of consequent commutation failures (or further commutation failures) and state how it can be minimised during operation and also at the design level.

[5]

Following a commutation failure for example from valve 1 to 3, valve 1 continues to conduct when valve 4 is fired and thereby, creates a direct short circuit across the inverter. Direct current shoots up due to shorting of the inverter. This increases the possibility of commutation failures in subsequent cycles due to increased current and is known as further/subsequent commutation failure.

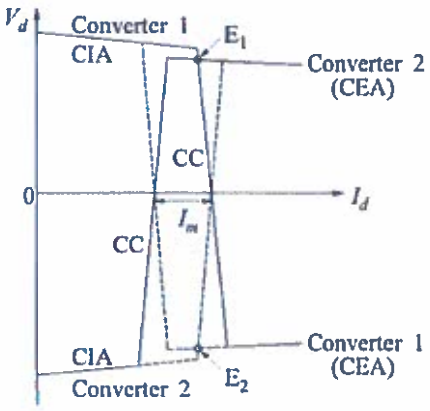
At an operational level the firing angle (β) is increased to prevent further commutation failure.

At a design level, higher value of smoothing reactor could be chosen (within other design trade-offs) to limit the rise of direct current and thus minimise the chances of further/consequent commutation failure.

[2 marks for explaining consequent commutation failure, 1.5 marks each for stating the remedy at operational and design level]

c) Explain, with the help of the rectifier and inverter control characteristics, how the power reversal is executed in a LCC HVDC link.

[5]



- Solid lines represent power transfer from converter 1 to 2
- Each converter is required to act as either rectifier or inverter and hence provided with a combined characteristics
- Solid lines represent power transfer from converter 1 to 2
- Reverse power flow for dotted line characteristics
- Power flow reversal is achieved by making the current order setting of converter 2 more than that of 1

[2 marks for correct control characteristics; 2 marks for explaining the power reversal process; 1 mark for proper reference to the control characteristics]

d) One end of a planned LCC HVDC link is to be connected at a 400 kV substation where the equivalent system impedance is 7.19 Ω. What should be the maximum rated capacity (in GW) of the HVDC link to ensure safe operation without the need for any fast voltage compensation? Assume that the reactive power consumption of the LCC is 55% of active power under the rated condition and an effective short-circuit ratio (ESCR) of 5 or more is required for safe operation.
[5]

Short circuit level at the substation is

$$S_{SC} = \frac{V_{rated}^2}{Z_{eq}} = 22253 \text{ MVA}$$

$$\text{ESCR} = \frac{S_{SC} - Q_c}{P_{DC}}$$

$$Q_c = -0.55 P_{DC}$$

$$P_{DC} = \frac{S_{SC}}{\text{ESCR} - 0.55}$$

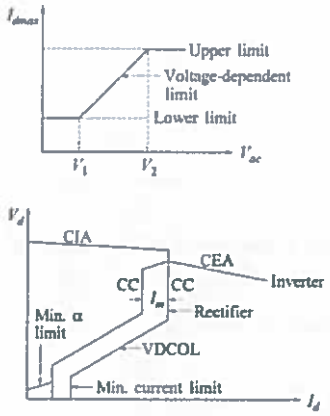
As it is required that for safe operation ESCR ≥5,

$$P_{DC} \leq \frac{22.253}{5 - 0.55} = 5 \text{ GW}$$

[1 mark for correct definition of ESCR, 1 mark for calculation of SCL, 3 marks for calculation of DC power]

2. a) State what is the voltage-dependant current order limit (VDCOL) and why is it necessary for LCC HVDC control? Sketch the converter control characteristics including VDCOL and explain its operation.

[5]



VDCOL reduces the current order when voltage (ac or dc) is below a threshold.

VDCOL is necessary as a fixed current order is not desirable under low voltage (inverter) condition due to the following reasons:

- reactive power demand increases due to increase in α
- reactive power supplied by the terminal filters and capacitors decreases with reduced voltage
- risk of commutation failure and voltage instability

[1 mark for VDCOL description, 3 marks for the need for VDCOL; 1 mark for sketch]

b) Explain the typical sequence of events following a fault within the AC system at the rectifier side of a LCC HVDC link. Assume appropriate control duty at both ends under the pre-fault condition.

[5]

- Under pre-fault condition, the rectifier end is in current control (CC) mode and the inverter end in extinction angle control (EAC) mode.
- During the rectifier side fault, the DC voltage at rectifier end and hence the current reduces.
- Current regulator decreases α to restore the current until it hits the limit before switching to constant firing angle mode
- Inverter switches to current control (CC) mode
- For large reduction in voltage, VDCOL come into play and might even have to shut the DC system down

[1 mark for each point]

c) Two separate 400 kV AC systems are interconnected by a 500 kV, 1.2 GW monopolar LCC HVDC link. The resistance of the DC line is 4 Ω and the commutation resistances at the rectifier and inverter ends are 6 Ω and 8 Ω, respectively. The minimum limit on the firing angle (α) is 4°. Under the normal condition with the rated voltage at the rectifier end, the link carries the rated current and operates with a firing angle (α) of 15°, extinction advance angle (γ) of 20° and 10% current margin. Due to a load event the rectifier side AC system voltage reduces to 85% of its normal value. Calculate the following under this reduced voltage condition. You may neglect the transformer tap changer action and activation of VDCOL.

(i) Reactive power consumption of the rectifier.

[6]

$$I_d = \frac{P_{dr}}{V_{dr}} = 2.4\ \text{kA}$$
$$V_{dor} = \frac{V_{dr} + R_{cr} I_d}{\cos\alpha} = 532.55\ \text{kV}$$

After AC side voltage reduction

$$V'_{dor} = 0.85 \times V_{dor} = 452.66\ \text{kV}$$

$$\cos\alpha' = \frac{V_{dr} + R_{cr} I_d}{V'_{dor}} > 1$$

Hence current control is transferred to inverter while the rectifier goes into constant firing angle (α_{min}) mode.

New current order for the inverter is
$I'_d = I_d - I_m = 2.16\ \text{kA}$

$$V'_{dr} = V'_{dor} \cos\alpha_{min} - R_{cr} I'_d = 438.6\ \text{kV}$$

Power factor at the rectifier end is

$$\cos\emptyset_r = \frac{V'_{dr}}{V'_{dor}} = 0.97$$

Reactive power consumption at the rectifier end is
$Q_r = V'_{dr} I'_d \tan\emptyset_r = 241.82\ \text{MVAr}$

[1 mark each for V_{dor}, mode-shift, I_d', V_{dr}', power factor and Q_r]

(ii) Reactive power consumption of the inverter.

[4]

$$V'_{di} = V'_{dr} - R_L I'_d = 429.96\ \text{kV}$$

Before the reduction in rectifier-side AC system voltage

$$V_{di} = V_{dr} - R_L I_d = 490.4\ \text{kV}$$
$$V_{doi} = \frac{V_{di} + R_{ci} I_d}{\cos\gamma} = 542.31\ \text{kV}$$

Power factor at the inverter end is

$$\cos\emptyset_i = \frac{V'_{di}}{V_{doi}} = 0.79$$

Reactive power consumption at the inverter end is
$Q_i = V'_{di} I'_d \tan\emptyset_i = 713.87\ \text{MVAr}$

[1 mark each for V_{di}', V_{doi}, power factor and Q_i]

Answers

3. a) Explain the variation of the reactive power consumed by the converter of a LCC HVDC link as the firing angle (α) is varied over the entire range covering the rectifier and the inverter operation. You may neglect the commutation overlap (μ).

[4]

Neglecting commutation overlap, it can be shown that the cosine of the firing angle (α) is equal to the cosine of the power factor angle (ϕ).

$\cos\alpha = \cos\varphi$

The range of variation of α is 0° to 180° before the commutation voltage reverses.

$0° < \alpha < 90°$ represents the rectifier operation with positive active power exchange while

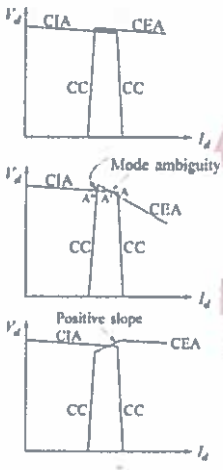
$90° < \alpha < 180°$ represents inversion operation with negative power exchange.

However, over this entire range $0° < \alpha < 180°$, the reactive power, which depends on $\sin\phi$, is positive. This implies that both rectifier and inverter consumes reactive power. The reactive power consumption increases with increase in firing angle (firing advance angle) of the rectifier (inverter).

[1 mark each for relationship between firing and power factor angle; rectifier/inverter operation; comment on reactive consumption and trends]

b) Explain the need for control mode stabilisation and state how it is usually achieved for control of LCC HVDC.

[5]



- Intersections of rectifier CIA and inverter CEA might not be well-defined for certain voltages
- Near the regions of intersection of CEA and CIA, there is a tendency of oscillation between multiple operating points A, A' and A''
- Control mode stabilisation is achieved by introducing a positive slope (constant β) at the junction of CC and CEA of inverters

[2 marks for explaining the problem; 1 mark for the solution; 1 mark for each figure]

c) State why one of the converter transformers used within a pole of a bipole LCC HVDC link use a star-delta (Y-Δ) configuration.

[3]

Star-delta connection creates a 30° phase difference between the line and phase quantities. This reduces the current harmonics on the AC side by creating a stair-case like waveform out of

rectangular current pulses for each phase. This also reduces the voltage harmonics on the DC side by creating 12 pulses out of the 6 pulses from individual converters.

[1 mark each for phase relationship, AC-side current harmonics and DC-side voltage harmonics]

d) A 1000 kV, 2.5 GW monopolar LCC-HVDC link is made up of 6-pulse converter bridges at both ends. The capacitor and filter banks installed at the inverter end of the link can provide reactive power support up to 60% of the rated active power of the link under normal AC system voltage. The DC line resistance is 5 Ω and the commutation resistance at either end is 4 Ω. The margin current setting is 8%. Calculate the maximum allowable extinction advance angle (γ) (in degrees) so that the reactive power demand of the inverter can be met locally by the capacitor and filter banks under the following conditions. Assume that the rated DC voltage is at the rectifier side and neglect any switching of capacitor and filter banks. Also assume that the inverter is in the current control mode under both conditions and maintains the same direct current. Consider appropriate reduction in reactive support due to reduction in AC system voltage at the inverter side.

(i) Rated power flow condition with normal AC system voltage at both ends.

[5]

As the inverter is under current control, the direct current is

$$I_d = \frac{P_{dr}}{V_{dr}} - I_m = 2.3 \text{ kA}$$

$$V_{di} = V_{dr} - R_L I_d = 988.5 \text{ kV}$$

$$P_i = V_{di} I_d = 2.27 \text{ GW}$$

$$\emptyset_i = \tan^{-1} \frac{0.6 \times P_{dr}}{P_i} = 0.583$$

$$V_{doi} = \frac{V_{di}}{\cos \emptyset_i} = 1184.3 \text{ kV}$$

Maximum extinction advance angle is

$$\gamma = \cos^{-1} \left(\frac{V_{di} + R_{ci} I_d}{V_{doi}} \right) = 32.6°$$

[1 mark each for I_d, V_{di}, ϕ_i, V_{doi} and γ]

(ii) Rated power flow condition but when the AC system voltage at the inverter end is 75% of that under normal condition.

[3]

Under reduced voltage condition the reactive power support available at the inverter end reduces in proportion to square of the voltage and is given by

$$Q'_{si} = 0.75^2 \times 0.6 \times P_{dr} = 1125 \text{ MVAr}$$

$$\emptyset'_i = \tan^{-1}\frac{Q'_{si}}{P_i} = 0.46$$

$$V_{doi} = \frac{V_{di}}{\cos \emptyset'_i} = 1102.9 \text{ kV}$$

Maximum extinction advance angle under reduced voltage condition is

$$\gamma' = \cos^{-1}\left(\frac{V_{di} + R_{ci}I_d}{V_{doi}}\right) = 25.23°$$

[1 mark each for Q'_{si}, V_{doi}, and γ']

Answers

Part B: Answer any 2 questions out of 3 from this part

4. Voltage Source Converters

a) Detail the main parts of a converter station, explaining their main purposes. Explain how these parts have changed between Voltage Source Converters (VSC) and Current Source Converters (CSC) and why.

[3]

A converter station mainly consists in (i) AC bus bars which interconnect the converter to the AC grid (it includes measurement sensors and protections), (ii) valve hall where all the power conversion takes place using powerful electronic devices, (iii) AC filter which smooth the current waveform out of the converter, (iv) AC transformers which adapt the voltage magnitude to the optimal converter operating point and provide galvanic isolation, (v) cooling system which outputs the heat generated by the power electronic devices and (vi) the technical room where both the control system is housed and the operators manage the station.
Since a VSC generates a much smooth current waveform than CSC, their AC filter footprint has shrunk by a significant amount but at the expense of bigger valve hall and cooling system due to the large volume and power losses inherent to the switch from thyristor to IGBT semiconductor technology.

[KEYWORDS]: AC bus bar, valve hall, AC filters, transformers, cooling system, technical room, AC current waveform, power losses
[1/2 mark for each points (i)-(vi)]

b) Using the electrical diagram of the ±200 kV DC multilevel Flying Capacitor converter depicted in Figure 4.1, complete the table using the format and example given below in order to answer this following questions:

i) What are the possible AC voltage levels?

[2]

[1/2 mark per voltage level for -200 kV, -100 kV, 100 kV and 200 kV]

ii) How many individual capacitors are conducting the AC current?

[3]

[1 mark per group of voltage level]

iii) What are the corresponding combinations of activated switches?

[3]

[1 mark per group of voltage level]

iv) Assuming that the AC current sign is positive, which semiconductor devices between the Diodes (Dx) and the Transistors (Tx) are carrying the current?

[2]

[1 mark for correct switch numbers and 1 mark for devices between D1, D2, D3, D4, T5, T6, T7, T8]

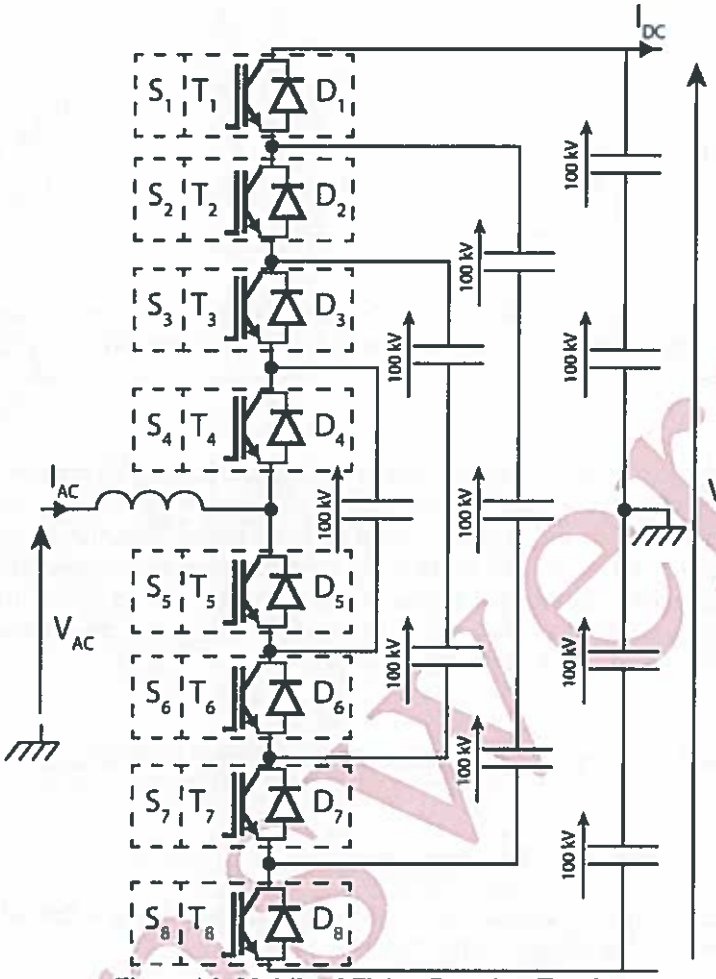


Figure 4.1: Multilevel Flying Capacitor Topology

Voltage Levels	Conducting Capacitors	Switch Combinations	Conduction Devices
0 kV	4	S1, S2, S5, S6	D1, D2, T5, T6
…	…	…	…

Voltage Levels	Conducting Capacitors	Switch Combinations	Conduction Devices
0 kV (+2-2)	4	S1, S2, S5, S6	D1, D2, T5, T6
200 kV (+2)	2	S1, S2, S3, S4	D1, D2, D3, D4
100 kV (+2-1)	3	S1, S2, S3, S5	D1, D2, D3, T5
100 kV (+2-2+1)	5	S1, S2, S4, S6	D1, D2, D4, T6
100 kV (+2-3+2)	7	S1, S3, S4, S7	D1, D3, D4, T7
100 kV (-2+3)	5	S2, S3, S4, S8	D2, D3, D4, T8
0 kV (+2-3+2-1)	8	S1, S3, S5, S7	D1, D3, T5, T7
0 kV (+2-3+1)	6	S1, S4, S6, S7	D1, D4, T6, T7
0 kV (-2+3-1)	6	S2, S3, S5, S8	D2, D3, T5, T8

0 kV (-2+3-2+1)	8	S2, S4, S6, S8	D2, D4, T6, T8
0 kV (-2+2)	4	S3, S4, S7, S8	D3, D4, T7, T8
-100 kV (+2-3)	5	S1, S5, S6, S7	D1, T5, T6, T7
-100 kV (-2+3-2)	7	S2, S5, S6, S8	D2, T5, T6, T8
-100 kV (-2+2-1)	5	S3, S5, S7, S8	D3, T5, T7, T8
-100 kV (-2+1)	3	S4, S6, S7, S8	D4, T6, T7, T8
-200 kV (-2)	2	S5, S6, S7, S8	T5, T6, T7, T8

c) Explain the reasons for the Flying Capacitor (FC) topology to be more suitable to use when the number of voltage levels is greater than 3 when compared to the Neutral Point Clamped (NPC) topology.

[3]

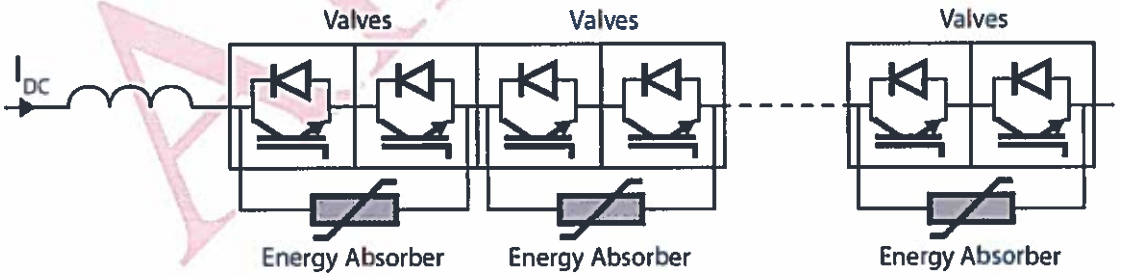
The number of split DC bus capacitors increases with the increasing number of voltage levels. Therefore, their voltage balancing becomes crucial to the good operation of the converter. In the case of the NPC, the switching pattern is fixed and depends only on the desired voltage level to be generated by the converter. This prevents the controller to properly adjust the voltage of these split DC bus capacitors. In the case of the FC, most of the voltage levels offer more than one possible switching combination, thus allowing the controller to alter the current path through the capacitors in order to prevent their voltage runaway.

[KEYWORDS]: increasing number of split capacitors, control flexibility, NPC = inflexible, FC = flexible.

[1 mark for each keyword concepts]

d) Explain, through a diagram, the components and operation of a full solid state DC circuit breaker. Comment on its strength and weaknesses.

[4]



The solid DC circuit breaker consists in a string of series IGBT modules in series in sufficient number such that the total voltage rating is at least equal to the DC bus voltage. During normal operation, all the IGBTs are conducting, thus acting like a low impedance dipole. In case of a DC side fault, all the IGBTs are blocked, increasing dramatically the impedance of the branch, quickly extinguishing the fault current. A DC inductor is usually connected in series with the solid state branch in order to reduce the ramp rate of the current to limit the peak of the fault current while the fault detection algorithm is reaching a decision.

This DC circuit breaker has the advantage of being very fast (<1 ms) but at the expense of having a large number of semiconductor devices in the conduction path during normal operation, thus increasing the overall converter power losses.

[KEYWORDS]: IGBTs in series, solid state switch, fast switching action, higher power losses.

[1 mark for each keyword concepts]

Answers

5. Modular Multilevel Converter

a) For an MMC, explain the following:

(i) The importance of the energy balancing.

[2]

The MMC relies on the charge in its submodules' capacitors to generate its voltage waveform. Since the arm current is periodically passing through these capacitors, they will exchange energy with the connected networks. To ensure good operation, the voltage level of these capacitors must thus be monitored and kept around their nominal value. This defines the purpose of the energy balancing controller in the MMC

[KEYWORDS]: charge in the submodule capacitor, keep the submodules' energy around their nominal value

[1 mark for each keyword concepts]

(ii) The difference between the horizontal and vertical energy balancing.

[3]

The energy mechanisms tackle different aspects of the energy balancing. The horizontal energy balancing exchanges energy between the phase leg (or phase unit), hence the horizontal name when referring to the electrical diagram of the MMC. This is achieved by circulating DC current between the legs. The vertical balancing exchanges energy between the top and bottom stacks in the same phase leg by running an AC current at the fundamental frequency. Since it is likely that each phase unit will require a different amount of balancing AC currents, additional quadrature AC currents are also circulated in order to cancel any remaining zero-phase sequence from leaking to the DC side.

[KEYWORDS]: Horizontal = between phases and uses DC current, vertical = between stacks and uses AC current, vertical balancing also involves quadrature AC currents to cancel zero-sequence currents to the DC bus.

[1mark for each keyword concepts: horizontal, vertical and zero-sequence cancellation]

b) A 800 MW MMC is connected to the latest DC cable technology rated a ±350 kV. Assuming that each half-bridge submodule is rated at 1.4 kV:

i) Compute a device count for the following: the number of IGBT modules, submodule capacitors.

[3]

Number of submodules per arm: $\frac{V_{DC}}{V_{cell}} = 500$

Total number of IGBT modules: $6 \times 2 \times N_{cell} = 6000\ IGBTs$

Total number of capacitors: $6 \times N_{cell} = 3000\ Capacitors$

[1 mark for each correct value]

ii) Estimate the minimal capacitance of the cell capacitors assuming that their voltage has to be kept within ±5% of their nominal value and the maximum peak-to-peak

energy deviation of a stack is 1.700 MJ for the specified operating envelope of this converter.

[3]

Cell Capacitor: $\frac{\Delta E}{2\, N_{cell}\, V_0^2\, \Delta V} = 17.3\ mF$

[1 mark for the correct equation, 2 marks for correct value]

iii) Calculate the total capacitive energy stored per unit power (provide the answer in kJ/MVA).

[3]

Energy per unit: $\frac{6\, N_{cell} \frac{C_{cell}}{2} V_{cell}^2}{S_{rated}} = 64\ kJ/MVA$

c) What are the motivations behind using mixed stacks in the MMC?

[6]

The classic MMC topology is based on the half-bridge submodules which offers the maximum power efficiency but has no negative voltage capability, thus making the resulting MMC weak to DC side fault. The opposite solution consists in replacing these half bridge submodules with full bridge submodules. However, this doubles the number of semiconductor devices in the conduction path and doubles the hardware expenses. A mitigated solution consists in mixing both half bridge and full bridge submodules in the stack, using the fact that, during DC side fault, the stack has only to provide plus to minus the AC peak voltage. And, since these values should never exceed half the DC bus voltage, the maximum negative voltage that the stacks have to generate is only half the DC bus voltage. Therefore, a mixed stack consisting of half of half-bridges and half of full-bridges should have both a limited increase in power losses while achieving full DC side fault blocking.

[KEYWORDS]: mix between full H-bridge and half-bridge submodule, 50% more losses than 2 half-bridges but 25% less than using only full H-bridge submodules, power losses mitigation.

[2 marks for keyword concept]

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6. Hybrid Voltage Source Converter

a) Using the general converter diagram shown in Figure 6.1:

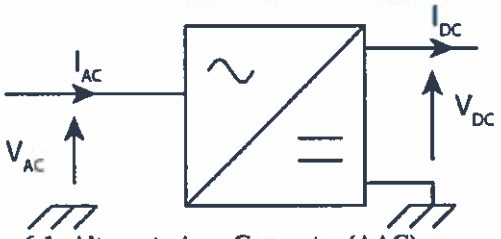


Figure 6.1: Alternate Arm Converter (AAC)

(i) Explain how the AC and DC quantities are related (assuming no power losses). [2]

A converter operates by processing the input power and sending it to the output port. In the case of an AC/DC converter, this means that the AC and DC quantities are linked by the following equation, where the average power from both sides equates:

$$< V_{AC} \times I_{AC} >=< V_{DC} \times I_{DC} >$$

[KEYWORDS]: AC=DC powers.

[2 mark for the concept clearly explained]

(ii) Furthermore, if the converter consists of stacks of submodules, what is the main operating principle which needs to be observed to ensure the proper operation of the converter? [3]

When using stacks of submodules inside the converter, particular attention must be brought in the operating principles of the converter. Since these submodules consist of a charged capacitor, the average energy deviation of the stacks must be zero over the course of a fundamental cycle. This implies that the energy level of any stacks at the end of the cycle must be the same as at the beginning of the cycle. This is imperative in order to ensure that the submodule capacitors do not overcharge or discharge over time.

[KEYWORDS]: charged capacitors, submodule capacitor voltage drift, average power of a stack = 0.

[1 mark for each keyword concept]

b) Consider the rudimentary converter made of 3 stacks as illustrated in Figure 6.2. The AC and DC quantities can be expressed by following equations:

$$V_{AC}(t) = V_{AC} \sin(\omega t)$$
$$I_{AC}(t) = I_{AC} \sin(\omega t)$$
$$V_{DC}(t) = V_{DC}$$
$$I_{DC}(t) = I_{DC}$$

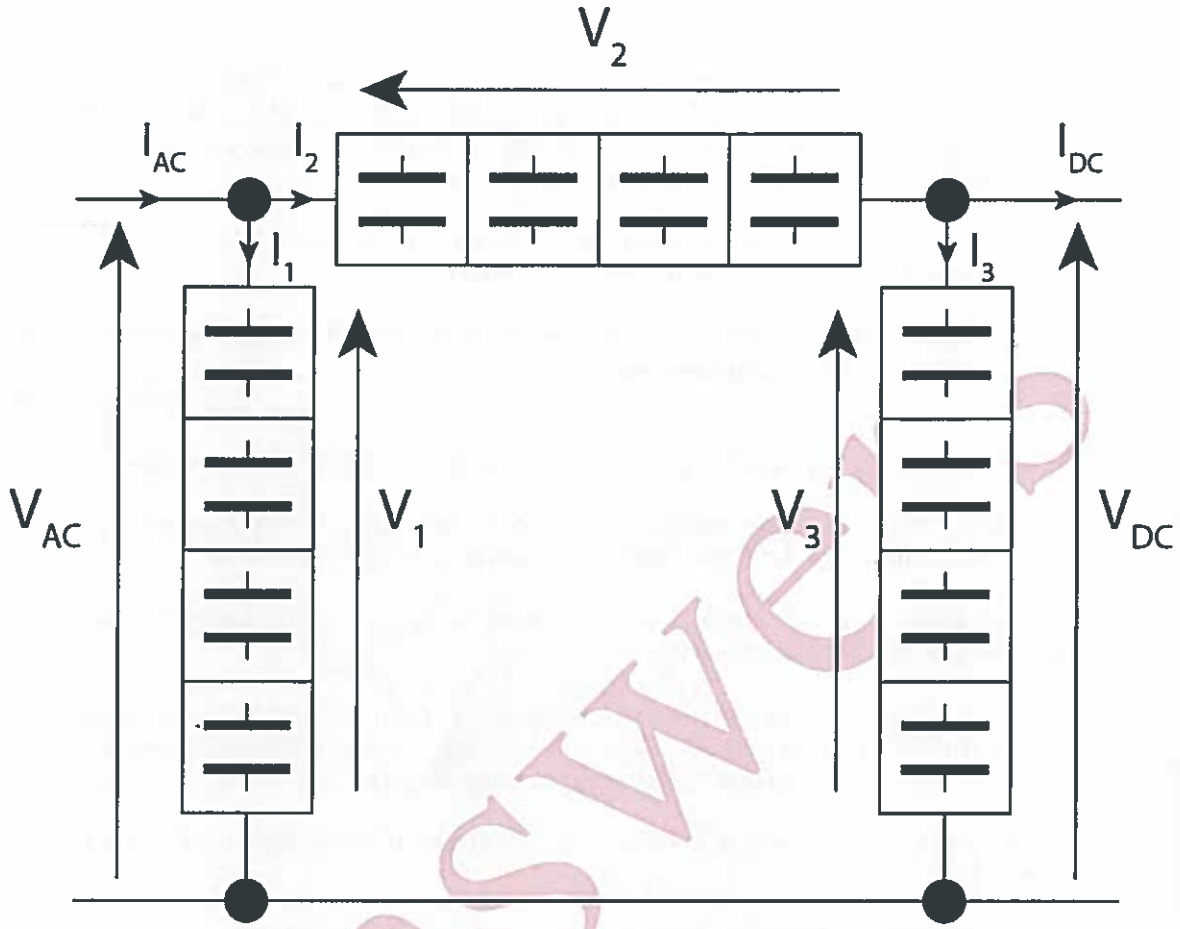


Figure 6.2: AAC as STATCOM

i) Derive the voltage waveform equations ($V_1(t)$, $V_2(t)$, $V_3(t)$) for Stacks 1, 2 and 3.

[3]

Using the voltage loop analysis, we have the following voltage equations:

$$V_1(t) = V_{AC}(t) = V_{AC}\sin(\omega t)$$
$$V_2(t) = V_{AC}(t) - V_{DC}(t) = V_{AC}\sin(\omega t) - V_{DC}$$
$$V_3(t) = V_{DC}(t) = V_{DC}$$

[1 mark per correct equation]

ii) If $V_{AC} = k \times V_{DC}$, state the maximum and minimum voltage characteristics of these stacks?

[3]

Using the relationship between the AC and DC voltage magnitudes, the stack voltage ranges are:

$$V_1 \in \{-k; k\} \times V_{DC}$$
$$V_2 \in \{-k-1; k-1\} \times V_{DC}$$
$$V_3 \in \{1; 1\} \times V_{DC}$$

[1 mark per correct equation]

iii) Infer what type of submodules can be used in order to minimize the power losses?

[3]

Regardless of the value of k, full-bridge submodules have to be used in Stack 1 and half-bridge submodules in Stack 3. Then, if k<1, then half bridge submodules can be used in stack 2, otherwise full-bridge or at best a mix of half- and full-bridge submodules have to be used in order to meet the alternative sign of the stack voltage waveform.

[1 mark for correct answers in both stack 1 and 3, 2 marks for correct answer in stack 2 with dependence on the modulation index]

iv) State the expression for the current waveforms in stacks 1, 2 and 3 to ensure proper operation of the converter? Justify your answers.

[6]

The converter will work properly only if the stacks' energy level stay constant over time.

Given that stack 1 has a voltage waveform equals to the AC voltage, then no in-phase AC current can be conducted through this stack. Therefore, its current waveform equation is:

$$I_1(t) = I_{DC}$$

The same applies to stack 3 which is in parallel with the DC bus. No DC current can be drawn from this stack, thus the current waveform of stack 3 is:

$$I_3(t) = I_{AC}(t) = I_{AC}\sin(\omega t)$$

Finally, stack 2 has a voltage waveform consisting of both AC and DC components. Using either the equality of AC to DC powers or the Kirchhoff's law, the current waveform for stack 3 is:

$$I_3(t) = I_{AC}(t) + I_{DC}(t) = I_{AC}\sin(\omega t) + I_{DC}$$

[KEYWORDS]: AC=DC powers, average power of a stack = 0, DC in stack 1, AC in stack 3 and mix in stack 2.

[1 mark per current equation and 1 mark per correct and clear justification]

Answers